

B.Tech. Degree VIII Semester Examination April 2016

EE 1804 E1 MECHATRONICS (2012 Scheme)

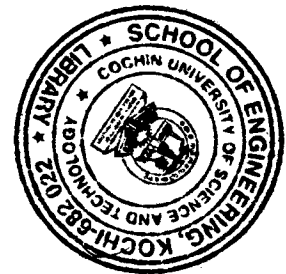
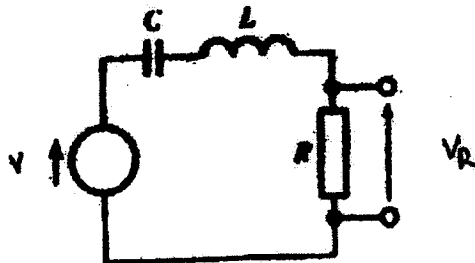
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What is mechatronics? Discuss on the scope of mechatronics.
 (b) Derive the relationship between the output, the potential difference across the Resistor R of V_R and the input V for the circuit shown.



- (c) Explain the principle of a pilot operated valve with the help of neat diagram.
 (d) State the parameters to be considered while selecting belt drives for particular application.
 (e) Explain open loop and closed loop in NC system.
 (f) Explain shortly on robot anatomy.
 (g) Explain derivative control. Why is it used along proportional controller?
 (h) Explain fibre optic devices in mechatronics.

PART B

(4 × 15 = 60)

- II. (a) Describe the key element of a mechatronics system. (5)
 (b) Explain the building blocks used for mechanical and electrical system. Relate their mathematical model. (10)

OR**(P.T.O.)**

- III. (a) Define the following performance measure for a second order system (5)
 (i) Rise time
 (ii) Peak time
 (iii) Overshoot
 (iv) Settling time
(b) Explain hydraulic-mechanical system. (10)
- IV. (a) Explain hydraulic and pneumatic power supply. (10)
 (b) Describe rotary actuators. (5)
- OR**
- V. (a) Explain belt drives. (5)
 (b) Explain stepper motor control. (10)
- VI. (a) List out the advantage of NC over other machining methods. (5)
 (b) Explain incremental and absolute systems. (10)
- OR**
- VII. (a) Explain different industrial robot configuration. (9)
 (b) Explain accuracy and repeatability. (6)
- VIII. Define MEMS. Explain various machining process to fabricate MEMS. (15)
- OR**
- IX. (a) What do you mean by PLC? Explain the architecture of a PLC. (10)
 (b) What are the criteria considered for selection of a PLC? (5)